Physics-Informed Neural Networks for Solving the DC Optimal Power Flow Problem

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***Abstract*—Physics-Informed Neural Networks (PINNs) provide a promising framework for learning optimization solutions while embedding domain-specific physical structure. This project develops a PINN-based approach for solving the DC Optimal Power Flow (DC-OPF) problem by jointly learning optimal generator dispatch and the associated dual variables directly from load measurements. The proposed model incorporates the Karush–Kuhn–Tucker (KKT) optimality conditions—including stationarity, power balance, generator and line-flow limits, dual feasibility, and complementary slackness—as differentiable penalty terms within the training objective. Using the IEEE 39-bus test system and a dataset of OPF solutions generated via YALMIP and GUROBI, we compare a purely supervised predictor with the proposed physics-informed model. Results demonstrate that embedding KKT-based physics dramatically improves feasibility: generator-capacity violation magnitudes decrease by $82.76\,\%$, line-flow violations by $76.41\,\%$, and power-balance mismatch by $56.49\,\%$. The PINN achieves these improvements while maintaining competitive accuracy in generator dispatch predictions. These findings highlight the potential of physics-informed learning to deliver fast, scalable, and constraint-aware approximations of OPF solutions for real-time power system applications.**

I. Introduction

Machine learning has emerged as a promising tool for accelerating computational problems in power system operations. Among these tasks, the Optimal Power Flow (OPF) problem plays a central role by determining the economically optimal dispatch of generators while satisfying network, equipment, and operational constraints. Classical OPF solvers, although accurate, may be too slow for applications involving repeated evaluations, such as scenario analysis, real-time control, or probabilistic simulations. As a result, significant research interest has shifted toward using supervised learning models to approximate optimal power flow (OPF) solutions.

A key challenge of ML-based OPF approximation is ensuring that predictions remain physically meaningful. Unlike optimization-based solutions, ML models do not inherently enforce power balance, thermal line ratings, or generator operational limits. Therefore, even highly accurate models may produce infeasible operating points that would be unacceptable in real-world grid operations. This motivates the need for systematic violation analysis to evaluate the reliability of ML predictions.

This project develops a complete workflow consisting of (i) a trained ML model for OPF variable prediction and (ii) a comprehensive constraint violation evaluation across a large test dataset. Three major OPF constraints are examined. First, generator capacity constraints ensure that predicted active power dispatch lies within allowable operating ranges. Second, transmission line flow constraints impose thermal limits on both the positive and negative directions of power transfer. Third, nodal power balance constraints guarantee that the sum of generation, demand, and line flows is physically consistent at each bus. For each constraint category, we compute the number of violations, total violation magnitude in MW, and average violation per sample. Tolerance thresholds are incorporated to avoid classifying small numerical deviations as violations.

The results provide detailed insights into model reliability, highlight conditions under which constraint violations occur, and motivate potential post-processing or physics-informed learning strategies. This analysis is essential for assessing whether ML-based OPF approximations can be safely integrated into real-world power system decision-making.

II. Related Work

The classical AC Optimal Power Flow (AC-OPF) problem is nonlinear and non-convex, making it computationally expensive for real-time and large-scale applications [1]. To reduce computational burden, the DC-OPF approximation is widely adopted as a linear reformulation that retains essential operational characteristics while improving solver performance [2]. Even under this simplification, evaluating OPF repeatedly under uncertainty or high-resolution simulations can remain costly.

Machine learning has emerged as a promising direction for accelerating OPF computation by learning mappings between system states and optimal dispatch decisions. Early works focused on supervised neural networks to identify optimal active constraint sets or approximate OPF solutions directly [3], [4]. While these models provide significant speed gains, one major limitation is that neural network predictions may violate generation limits, power balance constraints, or line thermal ratings, rendering solutions infeasible for physical systems.

To mitigate infeasibility, several constraint-aware deep learning approaches have been proposed. Pan et al. introduced DeepOPF as a regression model to approximate DC-OPF outputs [5], while Zhang et al. applied convex neural architectures to exploit the structural convexity of DC-OPF [6]. Other works integrated optimization principles into deep learning using primal-dual formulations or Lagrangian-based training [7], [8]. These methods improved constraint satisfaction compared to standard neural networks but still rely heavily on labeled datasets and may exhibit sensitivity to unseen operating conditions.

Physics-Informed Neural Networks (PINNs) offer an alternative paradigm by embedding governing physical laws into the model training process. Raissi et al. introduced the concept for solving partial differential equations using neural networks augmented with physics-based residuals [9]. Misyris et al. extended this approach to power systems, demonstrating that incorporating physical structure improves generalization and reduces dependency on large labeled datasets [10].

Recently, Nellikkath and Chatzivasileiadis applied PINNs to solve DC-OPF by embedding Karush–Kuhn–Tucker (KKT) optimality conditions into the loss function, enabling the neural network to learn both primal and dual variables while satisfying optimality constraints [11]. Their work additionally established worst-case violation guarantees using mixed-integer linear programming, demonstrating that PINNs outperform standard neural networks in constraint enforcement. Related verification frameworks such as those developed by Venzke et al. highlight the need for certifiable safety guarantees when machine learning models are used in operational decision-making [12].

Overall, the literature indicates a progression from supervised learning-based OPF approximators toward physically-constrained and certifiable deep learning models. Physics-informed OPF solvers, particularly those enforcing KKT consistency, represent a promising direction for scalable, reliable, and safe deployment of machine learning in power system optimization.

III. Methodology

This section presents the mathematical formulation of the DC Optimal Power Flow (DC-OPF) problem and the proposed Physics-Informed Neural Network (PINN) framework developed to learn optimal generator dispatch and associated dual variables directly from load measurements.

A. DC Optimal Power Flow

The DC-OPF problem is a linearized approximation of the AC-OPF problem. For a system with generator set $\mathcal{X}$, bus set $\mathcal{N}$, and transmission-line set $\mathcal{L}$, the DC-OPF is formulated as:

$$\min_{\{p_n\}_{n\in\mathcal{X}}} \quad \sum_{n\in\mathcal{X}} C_n(p_n) \tag{1}$$

$$\text{s.t.} \quad \sum_{n\in\mathcal{N}} (p_n - d_n) = 0, \tag{2}$$

$$\underline{p}_n \le p_n \le \bar{p}_n, \qquad \forall n \in \mathcal{X}, \tag{3}$$

$$\underline{o}_{l_{ij}} \le \sum_{n\in\mathcal{N}} \psi^n_{l_{ij}} (p_n - d_n) \le \bar{o}_{l_{ij}}, \qquad \forall\, l_{ij} \in \mathcal{L}. \tag{4}$$

Here, $C_n(p_n)$ denotes the generation cost at generator n, typically quadratic, and $\psi^n_{l_{ij}}$ is the Power Transfer Distribution Factor (PTDF) mapping a nodal injection at bus n to the flow on line l_{ij}. Each generator n has limits $\underline{p}_n$ and $\bar{p}_n$, and each line l_{ij} has flow limits $\underline{o}_{l_{ij}}$ and $\bar{o}_{l_{ij}}$.

Because the problem is convex, the optimal solution satisfies the Karush–Kuhn–Tucker (KKT) conditions. Introducing dual variables:

$$\lambda \text{ (power balance)},\ \underline{\tau}_n,\ \bar{\tau}_n \text{ (generator limits)},\ \underline{\mu}_{l_{ij}},\ \bar{\mu}_{l_{ij}} \text{ (line limits)},$$

The KKT system becomes:

1) Stationarity:

$$\frac{\partial C_n(p_n)}{\partial p_n} + \lambda - \underline{\tau}_n + \bar{\tau}_n + \sum_{l_{ij}\in\mathcal{L}} (\bar{\mu}_{l_{ij}} - \underline{\mu}_{l_{ij}}) \psi^n_{l_{ij}} = 0, \quad \forall n \in \mathcal{X}. \tag{5}$$

2) Primal Feasibility:

$$\sum_{n\in\mathcal{N}} (p_n - d_n) = 0, \tag{6}$$

$$\underline{p}_n \le p_n \le \bar{p}_n, \qquad \forall n \in \mathcal{X}, \tag{7}$$

$$\underline{o}_{l_{ij}} \le \sum_{n\in\mathcal{N}} \psi^n_{l_{ij}} (p_n - d_n) \le \bar{o}_{l_{ij}}, \qquad \forall l_{ij} \in \mathcal{L}. \tag{8}$$

3) Dual Feasibility:

$$\underline{\tau}_n,\ \bar{\tau}_n \ge 0, \quad \forall n \in \mathcal{X}, \qquad \underline{\mu}_{l_{ij}},\ \bar{\mu}_{l_{ij}} \ge 0, \quad \forall l_{ij} \in \mathcal{L}.$$

4) Complementary Slackness:

$$\lambda \left(\sum_{n\in\mathcal{N}} (p_n - d_n) \right) = 0, \tag{9}$$

$$\underline{\tau}_n (\underline{p}_n - p_n) = 0, \qquad \bar{\tau}_n (p_n - \bar{p}_n) = 0, \quad \forall n \in \mathcal{X}, \tag{10}$$

$$\bar{\mu}_{l_{ij}} \left(\sum_{n\in\mathcal{N}} \psi^n_{l_{ij}} (p_n - d_n) - \bar{o}_{l_{ij}} \right) = 0, \tag{11}$$

$$\underline{\mu}_{l_{ij}} \left(\underline{o}_{l_{ij}} - \sum_{n\in\mathcal{N}} \psi^n_{l_{ij}} (p_n - d_n) \right) = 0, \tag{12}$$

where the line flows are given by

$$f_{\ell_{ij}} = \sum_{n\in\mathcal{N}} \psi^n_{l_{ij}} (p_n - d_n).$$

These KKT conditions provide the physical constraints used in the construction of the physics-informed neural network.

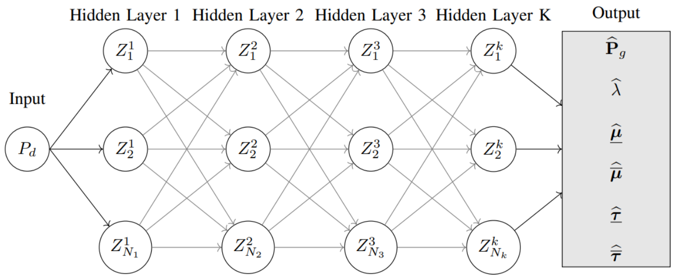

Fig. 1: Neural Network Architecture

B. Physics-Informed Neural Network Framework

The proposed PINN employs a fully connected feed-forward neural network with K hidden layers, illustrated in Fig. 1.

The layer outputs are defined recursively as

$$Z_{k+1} = \pi(W_{k+1} Z_k + b_{k+1}), \tag{12}$$

where W_{k+1} and b_{k+1} denote the weights and biases of layer $k+1$, respectively, and $\pi(\cdot)$ is the activation function.

For clarity, we introduce the internal affine transformation:

$$\hat{Z}_{k+1} = W_{k+1} Z_k + b_{k+1}, \tag{13}$$

$$Z_{k+1} = \tanh(\hat{Z}_{k+1}), \tag{14}$$

where $\tanh$ is used due to its smoothness and suitability for physics-informed models.

To enforce dual variable non-negativity for inequality constraints, we apply the rectified linear activation:

$$\pi_{\text{dual}}(x) = \max(0, x).$$

The network predicts the primal and dual OPF variables:

$$\left[\widehat{p}_n,\ \widehat{\lambda},\ \widehat{\underline{\tau}}_n,\ \widehat{\overline{\tau}}_n,\ \widehat{\underline{\mu}}_{l_{ij}},\ \widehat{\overline{\mu}}_{l_{ij}}\right].$$

Let $\mathcal{T}$ denote the training set with $N_t = |\mathcal{T}|$ samples. For any sample $k \in \mathcal{T}$, neural-network predictions are written as:

$$\hat{p}_n^{(k)}, \quad \hat{\lambda}^{(k)}, \quad \widehat{\underline{\tau}}_n^{(k)},\ \widehat{\overline{\tau}}_n^{(k)},\ \widehat{\underline{\mu}}_{l_{ij}}^{(k)},\ \widehat{\overline{\mu}}_{l_{ij}}^{(k)}.$$

Define per-sample residuals (KKT discrepancies) as follows:

1) Stationarity Residual:

$$r_{n,\text{stat}}^{(k)} = \frac{\partial C_n(\hat{p}_n^{(k)})}{\partial p_n} + \hat{\lambda}^{(k)} - \widehat{\underline{\tau}}_n^{(k)} + \widehat{\overline{\tau}}_n^{(k)} \tag{13}$$

$$+ \sum_{l_{ij} \in \mathcal{L}} (\widehat{\overline{\mu}}_{l_{ij}}^{(k)} - \widehat{\underline{\mu}}_{l_{ij}}^{(k)}) \psi_{l_{ij}}^n. \tag{14}$$

2) Primal Feasibility Residuals: Power Balance Violation

$$v_{\text{bal}}^{(k)} = \sum_{n \in \mathcal{N}} (\hat{p}_n^{(k)} - d_n^{(k)})$$

Generator bound violations:

$$v_{n,\ell}^{(k)} = \max\{0,\ \underline{p}_n - \hat{p}_n^{(k)}\}, \tag{15}$$

$$v_{n,u}^{(k)} = \max\{0,\ \hat{p}_n^{(k)} - \bar{p}_n\}. \tag{16}$$

Line flows and violations:

$$f_{l_{ij}}^{(k)} = \sum_{n \in \mathcal{N}} \psi_{l_{ij}}^n (\hat{p}_n^{(k)} - d_n^{(k)}), \tag{17}$$

$$v_{l_{ij},u}^{(k)} = \max\{0,\ f_{l_{ij}}^{(k)} - \bar{o}_{l_{ij}}\}, \tag{18}$$

$$v_{l_{ij},\ell}^{(k)} = \max\{0,\ \underline{o}_{l_{ij}} - f_{l_{ij}}^{(k)}\}. \tag{19}$$

3) Dual Feasibility Residual:

$$d_{\text{df}}^{(k)} = \sum_{n \in \mathcal{X}} \left(\max\{0, -\widehat{\underline{\tau}}_n^{(k)}\} + \max\{0, -\widehat{\overline{\tau}}_n^{(k)}\} \right) + \sum_{l_{ij} \in \mathcal{L}} \left(\max\{0, -\widehat{\underline{\mu}}_{l_{ij}}^{(k)}\} + \max\{0, -\widehat{\overline{\mu}}_{l_{ij}}^{(k)}\} \right). \tag{20}$$

4) Complementary Slackness Residual:

$$c_{\text{bal}}^{(k)} = \hat{\lambda}^{(k)} \cdot \sum_{n \in \mathcal{N}} (\hat{p}_n^{(k)} - d_n^{(k)}). \tag{21}$$

$$c_{n,\ell}^{(k)} = \widehat{\underline{\tau}}_n^{(k)} (\underline{p}_n - \hat{p}_n^{(k)}), \tag{22}$$

$$c_{n,u}^{(k)} = \widehat{\overline{\tau}}_n^{(k)} (\hat{p}_n^{(k)} - \bar{p}_n), \tag{23}$$

$$c_{l_{ij},u}^{(k)} = \widehat{\overline{\mu}}_{l_{ij}}^{(k)} (f_{l_{ij}}^{(k)} - \bar{o}_{l_{ij}}), \tag{24}$$

$$c_{l_{ij},\ell}^{(k)} = \widehat{\underline{\mu}}_{l_{ij}}^{(k)} (\underline{o}_{l_{ij}} - f_{l_{ij}}^{(k)}). \tag{25}$$

Mean-Absolute-Error (MAE) Discrepancies: All MAE terms average absolute residuals over the training set:

$$\text{MAE}_p = \frac{1}{N_t} \sum_{k \in \mathcal{T}} \sum_{n \in \mathcal{X}} \left| \hat{p}_n^{(k)} - p_n^{(k),\text{true}} \right|, \tag{26}$$

$$\text{MAE}_\lambda = \frac{1}{N_t} \sum_{k \in \mathcal{T}} \left| \hat{\lambda}^{(k)} - \lambda^{(k),\text{true}} \right|, \tag{27}$$

$$\text{MAE}_\tau = \frac{1}{N_t} \sum_{k \in \mathcal{T}} \sum_{n \in \mathcal{X}} \left(\left| \widehat{\underline{\tau}}_n^{(k)} - \underline{\tau}_n^{(k),\text{true}} \right| + \left| \widehat{\overline{\tau}}_n^{(k)} - \bar{\tau}_n^{(k),\text{true}} \right| \right), \tag{28}$$

$$\text{MAE}_\mu = \frac{1}{N_t} \sum_{k \in \mathcal{T}} \sum_{l_{ij} \in \mathcal{L}} \left(\left| \widehat{\underline{\mu}}_{l_{ij}}^{(k)} - \underline{\mu}_{l_{ij}}^{(k),\text{true}} \right| + \left| \widehat{\overline{\mu}}_{l_{ij}}^{(k)} - \bar{\mu}_{l_{ij}}^{(k),\text{true}} \right| \right), \tag{29}$$

$$\text{MAE}_{\text{stat}} = \frac{1}{N_t} \sum_{k \in \mathcal{T}} \sum_{n \in \mathcal{X}} \left| r_{n,\text{stat}}^{(k)} \right|, \tag{30}$$

$$\text{MAE}_{\text{pf}} = \frac{1}{N_t} \sum_{k \in \mathcal{T}} \left(\sum_{n \in \mathcal{X}} (v_{n,\ell}^{(k)} + v_{n,u}^{(k)}) + \sum_{l_{ij} \in \mathcal{L}} (v_{l_{ij},\ell}^{(k)} + v_{l_{ij},u}^{(k)}) + v_{\text{bal}}^{(k)} \right), \tag{31}$$

$$\text{MAE}_{\text{df}} = \frac{1}{N_t} \sum_{k \in \mathcal{T}} d_{\text{df}}^{(k)}, \tag{32}$$

$$\text{MAE}_{\text{cs}} = \frac{1}{N_t} \sum_{k \in \mathcal{T}} \left(\sum_{n \in \mathcal{X}} (|c_{n,\ell}^{(k)}| + |c_{n,u}^{(k)}|) + \sum_{l_{ij} \in \mathcal{L}} (|c_{l_{ij},\ell}^{(k)}| + |c_{l_{ij},u}^{(k)}|) + |c_{\text{bal}}^{(k)}| \right). \tag{33}$$

Final PINN Loss Function:

$$\mathcal{L}_{\text{PINN}} = \text{MAE}_p + \text{MAE}_\lambda + \text{MAE}_\tau + \text{MAE}_\mu + \alpha_{\text{stat}}\,\text{MAE}_{\text{stat}} + \alpha_{\text{pf}}\,\text{MAE}_{\text{pf}} + \alpha_{\text{df}}\,\text{MAE}_{\text{df}} + \alpha_{\text{cs}}\,\text{MAE}_{\text{cs}} \quad (34)$$

Here the scalar weights $\alpha_{\text{stat}}, \alpha_{\text{pf}}, \alpha_{\text{df}}, \alpha_{\text{cs}} > 0$ balance the relative importance of KKT components.

This formulation ensures that, when correctly trained, the PINN produces predictions consistent with the KKT conditions and thus with the true DC-OPF solution.

IV. Results and Analysis

This section presents the experimental setup and evaluation results for the proposed Physics-Informed Neural Network (PINN) model trained to approximate DC Optimal Power Flow (DC-OPF) dispatch and associated dual variables. The performance of the trained model is assessed in terms of prediction accuracy, primal feasibility, dual feasibility, and violation statistics across the unseen test dataset.

A. Simulation Setup

The dataset used for training and testing the model was generated using the IEEE 39-Bus New England test system. The Power Transfer Distribution Factor (PTDF) matrix was computed using MATPOWER, while optimal power flow solutions were obtained using the YALMIP modeling layer and the GUROBI solver in MATLAB. Load scenarios were generated using Latin Hypercube Sampling (LHS), allowing controlled perturbations of up to $\pm 20\%$ from the baseline bus demand. Only feasible OPF realizations were stored.

The final dataset consists of input samples corresponding to active power demand at load buses and output targets consisting of optimal generator dispatch, Lagrange multiplier for power balance, binding line constraint dual variables, and generator capacity shadow prices. Data preprocessing included normalization of inputs using the mean and standard deviation of the training subset.

Training and inference were performed using Google Colab with GPU acceleration enabled. The network architecture consists of a fully connected model with three hidden layers of 128 neurons each and `Tanh` activation functions. The model jointly predicts primal (P_g), equality dual variable λ, inequality duals for congestion (μ), and generator capacity multipliers (τ). Non-negativity of inequality dual variables was enforced using ReLU activation in the prediction layer.

The loss function is composed of supervised tracking terms and physics-informed components derived from Karush–Kuhn–Tucker (KKT) optimality conditions, including:

- Stationarity residual penalty
- Primal feasibility penalty (generator limits, line flow limits, and power balance)
- Complementary slackness consistency term
- Dual feasibility enforcement

This formulation ensures that the network not only approximates optimal decision variables but also respects feasibility and optimality structure, aligning with prior work in physics-informed optimization learning [8], [9], [11].

The network was trained for 2500 epochs using the Adam optimizer with a learning rate of 10^{-2} and mini-batches of size 64. Model weights achieving the lowest validation error were selected for final evaluation.

B. Physics-Informed Neural Network Performance Under Different Loss Weight Settings

To study the effect of embedding KKT-based physics penalties into the PINN training objective, we trained and evaluated the same neural architecture under two configurations:

1) **Case 1 – No physics-based penalties:** All physics weights are set to zero,

$$\alpha_{\text{stat}} = \alpha_{\text{pf}} = \alpha_{\text{df}} = \alpha_{\text{cs}} = 0,$$

so the network minimizes only supervised MAE terms. This acts as a baseline data-driven model.

2) **Case 2 – Physics-informed training:** Nonzero physics weights are used to penalize KKT discrepancies. The hyperparameters selected (after preliminary tuning to reduce worst-case generator constraint violations) are

$$\alpha_{\text{stat}} = 0.25, \quad \alpha_{\text{cs}} = 0.25, \quad \alpha_{\text{pf}} = 0.50, \quad \alpha_{\text{df}} = 0.25.$$

Both experiments used identical optimization settings to ensure a fair comparison: Adam optimizer, learning rate 1×10^{-2}, batch size 64, and 2500 epochs. The test set contains $N_{\text{test}} = 1996$ cases.

A. Training Behavior: The training-loss curves for the two cases are shown in Figures 2 and 3. Note that the Case 2 training loss is higher (for the same epoch budget) because the model must minimize additional physics residuals (KKT discrepancies) in addition to the supervised error; this typically increases the loss scale and the effective optimization difficulty.

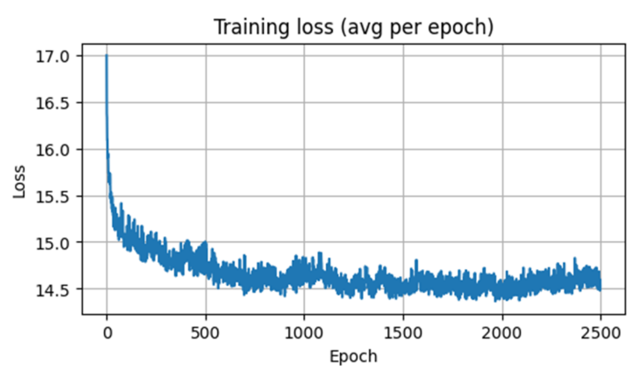

Fig. 2: Training loss (average per epoch): Case 1 (no physics penalties).

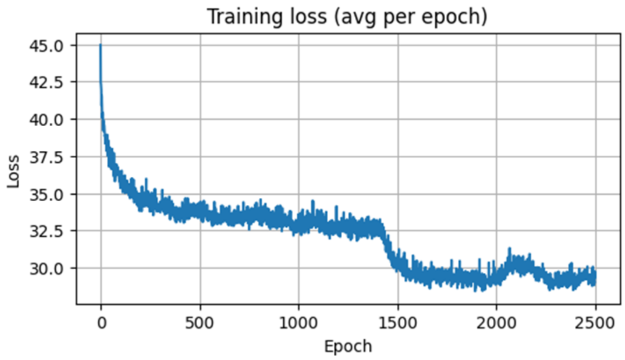

Fig. 3: Training loss (average per epoch): Case 2 (physics-informed).

B. Constraint Violation Statistics: Constraint violations were computed on the test set using the following tolerances:

- Generator and line constraints: 1% tolerance of the limit (i.e., a predicted value is considered violating only if it exceeds the limit by more than 1%),
- Power balance: absolute mismatch tolerance of 1 MW.

Tables I and II summarize the aggregated counts and MW magnitudes of violations for the two cases.

TABLE I: Constraint Violations (Test Set) — Case 1: All α's = 0

Violation Type	Count	Total MW	Average MW / sample
Generator capacity violations	24	564.81	0.28
Line flow violations	1750	50,515.97	25.31
Power balance violations	1956	86,928.65	45.35

TABLE II: Constraint Violations (Test Set) — Case 2: Nonzero physics weights

Violation Type	Count	Total MW	Average MW / samp
Generator capacity violations	6	97.40	0.05
Line flow violations	729	11,915.03	5.97
Power balance violations	1909	37,819.57	18.95

To quantify the improvement from Case 1 (all α values set to zero) to Case 2 (nonzero physics-based penalty weights), we compute the percentage reduction of each violation metric using

$$\text{reduction}(\%) = 100 \times \frac{\text{value}_{\text{Case 1}} - \text{value}_{\text{Case 2}}}{\text{value}_{\text{Case 1}}}.$$

The computed reductions are summarized below.

TABLE III: Percentage Reduction from Case 1 to Case 2 for All Violation Metrics

Violation Type	Count (%)	Total MW (%)	Avg MW/sample (%)
Generator Capacity	75.00%	82.76%	82.14%
Line Flow	58.34%	76.41%	76.41%
Power Balance	2.40%	56.49%	58.21%

These results indicate that, although the *number* of power-balance violations changes only slightly (a reduction of approximately 2.4%), the *magnitude* of the mismatch is reduced substantially (over 56% in total MW and more than 58% in average MW per sample). Generator-capacity and line-flow violations see particularly large improvements, with reductions exceeding 75% in MW terms. This demonstrates that introducing nonzero PINN penalty weights significantly enhances constraint satisfaction and improves overall model feasibility.

C. Generator Dispatch Comparison: Figure 4 and Figure 5 compare the average generator dispatch over all test samples between the PINN predictions and ground-truth OPF solutions for Case 1 and Case 2, respectively.

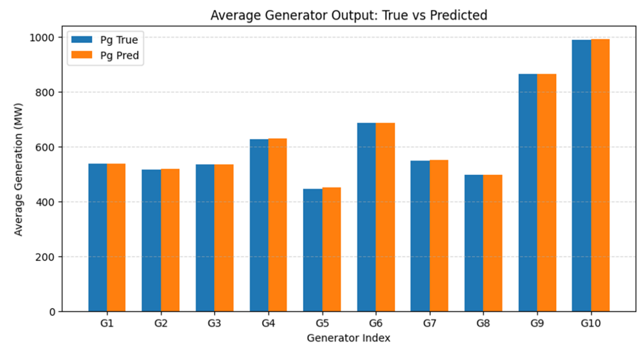

Fig. 4: Average generator dispatch: Case 1 (no physics penalties).

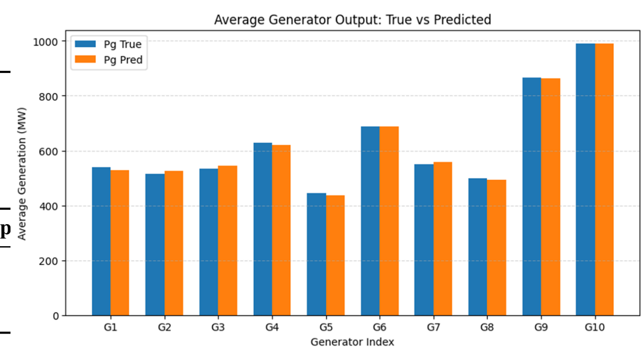

Fig. 5: Average generator dispatch: Case 2 (physics-informed).

D. Key Observations:

- Embedding KKT-derived physics terms (Case 2) dramatically reduces *violation magnitudes* for generator and line constraints (over 76% reduction in total MW for lines and over 82% for generator capacity), and reduces counts as well.
- Power-balance violation counts are similar between the two cases, but the magnitudes are much smaller under physics-informed training; this indicates the PINN reduces the severity of mismatches even when residual sign/occurrence persists for borderline samples.
- Training loss is larger in Case 2 for the same epoch budget, reflecting the additional physics residuals the

optimizer must minimize. This suggests that training schedules or stronger optimizers / longer training may further improve physics enforcement without sacrificing predictive quality.

V. Discussion and Conclusion

A. Discussion

The experiments above compare a purely supervised neural predictor (Case 1) with a physics-informed PINN that explicitly enforces KKT optimality conditions (Case 2). The results support several important conclusions:

1) **Effectiveness of physics-based penalties.** Introducing stationarity, feasibility, complementary slackness, and dual-feasibility penalties yields substantial reductions in both the *magnitude* and *frequency* of constraint violations. This is especially pronounced for generator capacity and line-flow constraints where the total MW of violations decreased by roughly 82.8% and 76.4%, respectively.
2) **Trade-off between pointwise accuracy and feasibility.** Case 1 (no physics) may optimize MAE on supervised labels more aggressively, but at the expense of feasibility. Case 2 trades some supervised error to satisfy physical constraints and to reduce worst-case violations which is a desirable property for operational deployment.
3) **Power balance behavior.** The number of power-balance violation events changed only slightly, but the violation magnitudes were greatly reduced. This suggests that the PINN is effective at shrinking mismatches even when rare sign switches or boundary cases remain.
4) **Training complexity.** Adding KKT-based penalties increases the effective difficulty of the optimization problem. The training loss and training time typically increase for the same number of epochs. Remedies include longer training, annealing schedules for α's, adaptive weight-tuning (e.g., uncertainty-based weighting), or specialized optimizers.
5) **Hyperparameter sensitivity.** The specific alpha weights chosen ($\alpha_{\rm stat} = 0.25,\ \alpha_{\rm pf} = 0.5,\ \alpha_{\rm df} = 0.25,\ \alpha_{\rm cs} = 0.25$) were selected after limited tuning to reduce worst-case generation violations. A more systematic hyperparameter search could further improve the trade-off between accuracy and feasibility.

B. Limitations

- **Dataset dependency.** The PINN performance depends on the representativeness of the training data: out-of-distribution loads may lead to larger violations.
- **Scalability.** For very large networks (many buses/lines/generators), the dimension of the PTDF and the number of dual outputs grow, increasing network size and training cost.

C. Conclusion and Future Work

This work demonstrates that a single, unified PINN that jointly predicts primal and dual variables and minimizes an objective embedding KKT residuals can substantially reduce constraint violations relative to a purely supervised predictor. Key takeaways:

- Physics-informed training yields large reductions in violation *magnitudes* (up to $\sim 83\%$ for generator-capacity MWs and $\sim 76\%$ for line-flow MWs in our experiments).
- The PINN approach improves operational feasibility while retaining competitive prediction accuracy.
- There exists an expected training cost trade-off; increasing epochs or using adaptive weight strategies can mitigate this.

a) Recommended future directions:

1) **Adaptive weighting of physics terms.** Use dynamic weight adjustment (learning weights, uncertainty weighting) to balance supervised vs physics losses automatically.
2) **Longer/advanced training.** Train with larger epoch budgets and/or advanced optimizers (AdamW, LAMB) and learning-rate schedules to further reduce KKT residuals.
3) **Robustness testing.** Evaluate PINN performance on out-of-distribution load scenarios and under contingency (N-1) conditions.
4) **Scalability.** Apply the approach to larger realistic networks and report runtime/memory trade-offs.
5) **Hybrid deployment.** Combine the PINN as a fast warm-start for a solver (e.g., use predictions as an initial point for an OPF solver), enabling both speed and guaranteed feasibility.

Overall, the PINN approach provides a promising route to fast, approximately optimal OPF predictions with significantly improved feasibility compared to purely supervised models. The empirical reductions in violation magnitudes illustrate the value of enforcing physical (KKT) structure during training.

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